

Advanced (Habanero)

Q1. Simplify the expression $\sqrt{16}$

- (A) 2
- (B) 4
- (C) 6
- (D) 8
- (E) 10

Q2. What is the value of $\sqrt{25}$?

- (A) 3
- (B) 5
- (C) 7
- (D) 9
- (E) 11

Q3. Simplify $\sqrt{64}$

- (A) 6
- (B) 8
- (C) 10
- (D) 12
- (E) 14

Q4. Convert $\sqrt{x}$ to rational exponent form

- (A) $x^{1/3}$
- (B) $x^{1/2}$
- (C) $x^{2/3}$
- (D) $x^{3/4}$
- (E) $x^{4/5}$

Q5. What is the value of $\sqrt[3]{27}$?

- (A) 2
- (B) 3
- (C) 4
- (D) 5
- (E) 6

Q6. Simplify the expression $\sqrt{49}$

- (A) 5
- (B) 6
- (C) 7
- (D) 8
- (E) 9

Q7. Convert $x^{1/2}$ to radical form

- (A) $\sqrt[3]{x}$
- (B) $\sqrt{x}$
- (C) $\sqrt[4]{x}$
- (D) $\sqrt[5]{x}$
- (E) $\sqrt[6]{x}$

Q8. Simplify the expression $\sqrt[3]{125}$

- (A) 4
- (B) 5
- (C) 6
- (D) 7
- (E) 8

Open Ended Questions (FRQ)

Q9. Simplify the expression $\sqrt{48}$ and show your work

Q10. Convert $\sqrt[4]{x}$ to rational exponent form and simplify $\sqrt{75}$

Chef's Tips

- Tip 1: Make sure to check for any common factors before simplifying radicals
- Tip 2: Pay close attention to the index of the radical when converting to rational exponent form
- Tip 3: Simplify all radicals before adding or subtracting